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Project Title:	Mechanisms of Skeletal Muscle Atrophy Due to Simulated Weightlessness		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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Task Description:	It is well known that exposure to spaceflight leads to marked decreases in skeletal muscle size and functional capacity, but the mechanisms underlying these changes are poorly understood. During the previous 3 yrs of NSBRI support, our lab has completed an extensive microarray analysis of soleus muscles over a time course of unloading. These data will be used as a source of candidate genes to test for regulatory roles in the atrophy process using genetic approaches in muscle cell culture and in whole muscle. The overall aim of this project is to identify genes involved in the regulation of protein loss during muscle atrophy. In addition we will test drugs of the NSAID family that have been suggested to have anti-atrophy effects. We will also test whether specific drugs that inhibit proteins involved in NF-kappaB signaling, which are known to be required for atrophy, can inhibit atrophy due to simulated weightlessness. These drugs include, helenalin, andrographolide, and wedelolactone. In work to identify genes that are sufficient to induce atrophy we will use genes from the microarray analysis that were activated early with unloading, had sustained activation, and are involved with protein degradation. These genes include cathepsins C, D, and L, the ubiquitin ligases Nedd4 and atrogin-1, and calpastatin. As atrophy also involves decreased protein synthesis, we will also study IGFBP5, a gene involved in the negative growth control of IGF-1. As endpoint markers of atrophy we will measure cell diameter and DNA/protein ratio, but we will also measure changes that represent intermediate steps in the atrophy process. These include the phosphorylation state of proteins involved in translation, the activity of proteolytic systems, and protein synthetic and degradation rates. In addition, co-expression strategies will be used to test whether multiple genes more effectively induce markers of atrophy. Where possible we will also test whether the inhibition of candidate genes can attenuate atrophy in a myotube atrophy model, thus testing whether the candidate is necessary for atrophy in cell culture. To test if a candidate gene is necessary for unloading induced atrophy, we will inhibit its expression by injecting a dominant negative or an anti-sense version of the candidate gene contained in plasmid or viral expression vectors. We will then measure if unloading atrophy can be attenuated in fibers containing the transgene. Uptake by individual muscle fibers will be tracked using epitope-tagged cDNAs or EGFP fusion proteins in order to colocalize overexpression with the effects on fiber size. Knockout mice will also be used to evalute the ability of candidate genes to attenuate unloading muscle atrophy. By identifying genes involved in atrophy we will be in a better position to develop more effective nutritional or pharmacological countermeasures to combat the deleterious changes in muscle function due to exposure to spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	By identifying genes involved in atrophy we will be in a better position to develop more effective nutritional or pharmacological countermeasures to combat the deleterious changes in muscle function due to any type of disuse atrophy.
Task Progress:	None of the data below are yet published but at least half of these data have been presented in abstract form at scientific meetings. Part I: Effect of drug administration on blocking atrophy due to simulated weightlessness. Certain NSAIDS have been shown to inhibit atrophy due to cachexia at high doses. Therefore we carried out studies to test several NSAIDS as well as other traditional medicines with wide safety profiles to determine if atrophy could be inhibited. The use of NSAIDS has been used since they are known to inhibit some proteins that are involved in atrophy such as prostaglandins and NF-kappaB. In our hands, neither a high (300mg/kg, 2x per day) nor a low (187 mg/kg, 2x per day) dose of salicylate inhibited atrophy or NF-kappaB activity due to 7 days of simulated weightlessness in rat hindlimb muscles (the soleus and the plantaris). In addition, naproxen given twice a day (20mg/kg/day) also did not block unloading induced atrophy or NF-kappaB activity. There was also no effect with high dose curcumin (100mg/kg/day). On the other hand aspirin administered twice a day at 30mg/kg (i.e., 60mg/kg/day) showed a 40% attenuation of atrophy without a change in NF-kappaB activity. All these studies used intraperitoneal injections of the drugs. A drawback of these studies, even the ones showing atrophy inhibiting effects of the drugs, are that the drug doses are not well tolerated by humans, even when normalized for their lower metabolic rates (per kg body mass). Therefore in ongoing work we are using traditional medicines that have a wide safety margin. In recent exciting work, we have found that andrographolide, a medicine that inhibits the NF-kappaB family member that is required for atrophy in rodents called p50, inhibits atrophy when the drug is mixed with chow and the rats ingest 20mg/kg/day. At this dose the inhibition of muscle fiber atrophy due to 7 days of simulated weightlessness was 25%, with the drug plus chow mixture being given for a total of 8 days; 1 day before unloading plus the 7 days of unloading (weightlessness). However, we will test a higher dose in the next series of studies; a dose that will also be safe for use in humans. We also have planned studies using helenalin and wedelolactone, other safe traditional medicines, which inhibit other components of NF-kappaB signaling. In the next two paragraphs (Part II and III) of this TaskBook update we describe experiments that identify more specifically the cellular components that are responsible for muscle atrophy. This work is essential to continue to improve upon efforts toward developing effective countermeasures. Part II: The ubiquitin-protein ligase Nedd4 regulates Notch1 in tension sensitive models of skeletal muscle atrophy. Skeletal muscle atrophy results from muscular inactivity as well as from disease states, and is characterized by enhanced ubiquitination-dependent proteolysis of muscle proteins. Several muscle specific ubiquitin-protein ligases have been shown to be involved in this process. Nedd4, a ubiquitously expressed HECT-domain containing ubiquitin-protein ligase, showed increased mRNA expression in rat soleus muscles after 1, 4, 7, and 14 days of hind limb unloading. Nedd4 protein localized to the sub-sarcolemmal region of muscle fibers. Hind limb unloading, sciatic denervation, food deprivation, and diabetes all led to atrophy of soleus, plantaris, and gastrocnemius muscles, but only unloaded or denervated muscles showed an increase in Nedd4 protein expression. This increase was correlated with decreased levels of Notch1 receptor, a known target of Nedd4. An inhibition of the unloading-induced downregulation of Notch1 was seen when a dominant negative form of Nedd4 (Nedd4(c to s substitution)) was injected into muscles prior to unloading, indicating that Nedd4 is required for the downregulation of Notch1. Furthermore, overexpression of wild type Nedd4 in soleus muscles of weight-bearing rats caused a decrease in Notch1 protein. Finally, conditional overexpression of Nedd4 in C2C12 myotubes induced ubiquitination of Notch1. Thus Nedd4 regulates Notch1 receptor signaling in tension-dependent muscle atrophies, but not in disease atrophy, indicating a role for Notch1 signaling in the inactivity atrophy phenotype. Part III: IkappaBalpha but not c-Rel, is required for skeletal muscle atrophy.

	Skeletal muscle atrophy is associated with a marked and sustained activation of NF-kappaB activity. Previous work showed that p50 is one of the NF-kB (i.e., Rel) family members required for this activation and for muscle atrophy. In this work we tested whether another NF-kB family member, c-Rel, is required for atrophy, and, we tested if the inhibitor of kappaB (IkappaBalpha), which binds and retains Rel proteins in the cytosol, is required for atrophy. Endogenous IkBa was activated (i.e., decreased) at 3 and 7 days of muscle disuse (i.e., hind limb unloading). Therefore, a dominant negative IkBa (IkBa-deltaN), which is resistant to degradation, was electrotransfered into soleus muscles to determine if IkBa degradation is required for unloading-induced atrophy and its associated biochemical markers. IkBa-deltaN expression abolished both NF-kB activation and the increase in total ubiquitinated protein due to muscle unloading. IkBa-deltaN inhibited unloading-induced fiber atrophy by almost one-half. The expression of certain genes known to be upregulated with atrophy were significantly inhibited by IkBa-deltaN expression during unloading, including MAFbx/atrogin-1, Nedd4, IEX, 4E-BP1, FOXO3a, and cathepsin L, suggesting these genes may be targets of NF-kB transcription factors. In contrast, c-Rel was not required for atrophy because the unloading-induced decrease in fiber area, increase in ubiquitinated protein, and activation of NF-kB activity were the same in c-Rel-/- and wild type mice. Thus, IkBa degradation is required for the unloading-induced decrease in fiber size, the increase in protein ubiquitination, activation of NF-kB signaling, and the expression of specific atrophy genes, but c-Rel is not. These data identify additional IkB and NF-kB family members that are either essential or nonessential for skeletal muscle atrophy and provide candidate NF-kB target genes for further study.
Bibliography Type:	Description: (Last Updated: 09/05/2007)
Articles in Peer-reviewed Journals	Kandarian SC, Jackman RW. "Intracellular signaling during skeletal muscle atrophy. " Muscle Nerve. 2006 Feb;33(2):155-65. PMID: 16228971 , Feb-2006
Articles in Peer-reviewed Journals	Stevenson EJ, Koncarevic A, Giresi PG, Jackman RW, Kandarian SC. "Transcriptional profile of a myotube starvation model of atrophy." J Appl Physiol. 2005 Apr;98(4):1396-406. Epub 2004 Dec 17. PMID: 15608089 , Apr-2005
Articles in Peer-reviewed Journals	Hunter RB, Kandarian SC. "Disruption of either the Nfkb1 or the Bcl3 gene inhibits skeletal muscle atrophy." J Clin Invest. 2004 Nov;114(10):1504-11. PMID: 15546001 , Nov-2004
Books/Book Chapters	Kandarian SC. "Large scale gene expression profiles as tools to study skeletal muscle adaptation." in "Skeletal muscle plasticity in health and disease : from genes to whole muscle." Ed. R. Bottinelli and C. Reggiani. Dordrecht : Springer, c2006. (Advances in muscle research ; v. 2), Jul-2006
Papers from Meeting Proceedings	Judge A, Koncarevic A, Kandarian S. "Inhibition of IkB alpha degradation prevents NF-kB activation and attenuates fiber atrophy and gene expression during unloading. This NSBRI post-doc won a Young Investigator Award for this work." FASEB Summer Research Conference, Skeletal Muscle Satellite and Stem Cells, Tucson, AZ. June 11-16, 2005. FASEB Summer Research Conference, listed in meeting program, June 2005. , Jun-2005
Papers from Meeting Proceedings	Van Gammeren D, Judge A, Kandarian SC. "The role of IKKbeta and IKKalpha in the activation of NF-kB during skeletal muscle atrophy." Frontiers in Myogenesis Meeting, Callaway Gardens, Pine Mountain, Georgia, April 27 - 30, 2006. Frontiers in Myogenesis, Abstract in meeting program, April 2006. , Apr-2006
Papers from Meeting Proceedings	Kandarian S. "Signaling mechanism in skeletal muscle fiber atrophy." 2006 Experimental Biology, Symposium on Cellular Signals Regulating Skeletal Muscle Type and Size, San Francisco, CA, April 1-5, 2006. 2006 Experimental Biology, Symposium #184, April 2006. , Apr-2006
Papers from Meeting Proceedings	Jackman RW, Hunter RB, Kandarian SC. "Changes in cultured myotubes mediated by the adenoviral delivery of an EGFP-Bcl-3 fusion protein." NASA's Bioastronautics Investigators' Workshop, Galveston, TX, January 9-12, 2005. NASA's Bioastronautics Investigators' Workshop, January 2005. , Jan-2005